Nonlinear MPC Controls for an Autonomous Underwater Vehicle Simulation*

Maggie Ni[1] and Mo Phoompho[2]

Abstract— **This work presents a nonlinear Model Predictive Control (NMPC) approach for trajectory tracking and depth regulation of a BlueROV2 underwater vehicle. A high-fidelity Gazebo simulation is used to model the vehicle's nonlinear dynamics, sensors, and actuator constraints. The NMPC controller solves a constrained finite-horizon optimization problem in a receding horizon framework. Closed-loop simulations demonstrate effective tracking performance and robustness under disturbances, noise, and actuator limitations, highlighting the suitability of NMPC for underwater vehicle control.**

I. INTRODUCTION

Autonomous and remotely operated underwater vehicles (AUVs and ROVs) play a critical role in marine exploration, infrastructure inspection, and environmental monitoring. However, the development and validation of reliable control strategies for these systems remain challenging due to the complex, nonlinear hydrodynamic interactions, limited sensing capabilities, and constrained operating environments inherent to underwater domains. Experimental testing in real-world conditions is often costly, time-consuming, and difficult to reproduce, motivating the use of high-fidelity simulation environments for controller design and evaluation.

Model Predictive Control (MPC) has emerged as a powerful framework for controlling nonlinear systems with constraints, offering the ability to explicitly account for system dynamics, actuator limitations, and performance objectives within an optimization-based formulation. In underwater applications, MPC is particularly attractive due to its capability to handle multivariable dynamics and enforce constraints critical for safe and efficient operation.

In this work, a nonlinear Model Predictive Control (NMPC) strategy is developed and evaluated for an underwater robotic platform within a simulation-based framework. The objective is to achieve accurate trajectory tracking and depth regulation while accounting for the nonlinear behavior and physical constraints of the system. To enable systematic development and testing, a high-fidelity simulation environment is employed, allowing for repeatable experiments under controlled conditions.

II. RELATED WORKS

A. Control Architectures

Various control architectures have been proposed to improve the stability and robustness of AUV navigation during pipeline inspections. [1] presents a modular control architecture based on a high-fidelity 6-DOF model of the BlueROV2 that captures vehicle dynamics, hydrodynamics, thruster behavior, ocean currents, and tether effects. Within this framework, sliding-mode control generates body-frame forces and moments that are mapped to individual thruster commands through a thruster-allocation module.

Akram and Casavola [2] and Xia et al. [3] both use feedback control to maintain vehicle alignment with linear underwater structures. [2] employs a vision-based control scheme that extracts image features such as lateral offset and heading error to guide pipeline tracking, while [3] uses a two-layer architecture where a heading-planning module generates desired headings that are tracked by a feedback controller.

The use of model predictive control for AUV is proposed in [4], where it is integrated with dynamic local path planning. A spline-based planner generates reference trajectories, and a nonlinear MPC controller tracks them through repeated receding-horizon optimization while considering the vehicle's nonlinear dynamics and depth control requirements. This approach supports adaptive navigation under limited sensing information and uncertain environments. [5] focuses specifically on the implementation of MPC for BlueROV2. Experimental results indicate that the implemented MPC provides effective closed-loop control and outperforms PID and LQR baselines.

Although sliding-mode and feedback-based control methods have shown promising results, they present several limitations for the present application. Sliding-mode control may lead to chattering and less smooth actuator commands, while vision- or heading-based feedback control depends strongly on the quality of sensed features and typically reacts to tracking errors rather than predicting future vehicle behavior. By comparison, MPC provides a predictive and constraint-aware framework that can explicitly account for system dynamics, optimize control actions over a receding horizon, and better handle the coupled motion of the BlueROV2. Hence, MPC is selected in this study as the primary control strategy for pipeline inspection.

B. Simulation Environment

Simulation environments are commonly used to develop and test underwater robotics algorithms before real-world deployment. The AUV Simulator provides a ROS–Gazebo framework that models underwater vehicle dynamics, sensors, thrusters, and environmental effects such as hydrodynamic forces and water currents [6]. Using this platform,

*This work was not supported by any organization

[1]Maggie Ni is with Department of Aerospace, University of Urbana-Champaign, Champaign County, IL, USA

[2]Mo Phoompho is with Department of Aerospace, University of Urbana-Champaign, Champaign County, IL, USA

[7] constructs a closed-loop AUV pipeline inspection scenario where Gazebo simulates the underwater environment, vehicle dynamics, and pipeline geometry while providing virtual sensors such as cameras, sonar, and chemical plume detectors. ROS nodes process sensor data, estimate vehicle state using an EKF, compute navigation errors relative to the pipeline, and apply a fuzzy-logic controller to generate thrust commands, allowing the simulated AUV to detect, follow, and inspect pipelines within a realistic virtual environment.

The Angler framework [8] demonstrates the potential of ROS2, Gazebo, and ArduSub SITL for underwater simulation by providing a modular environment for closed-loop testing, motion planning, and controller integration. It also supports benchmarking in procedurally generated environments with static and dynamic obstacles. From these promising results, this study adopts Orca4 [9], which uses similar simulation architecture but is more specifically designed for BlueROV2 platform.

III. PROBLEM STATEMENT

Accurate navigation and depth regulation of autonomous underwater vehicles remain challenging due to highly nonlinear vehicle dynamics, hydrodynamic disturbances, actuator limitations, and the strong coupling between translational and rotational motion. Conventional control methods such as PID are simple to implement but may struggle to achieve high tracking accuracy under these nonlinear operating conditions, particularly when system constraints and multivariable interactions must be considered simultaneously. This motivates the investigation of advanced model-based control strategies for underwater vehicle operation.

In this work, the BlueROV2 is selected as the experimental platform because of its six-thruster vectored configuration, maneuverability, and widespread use in underwater robotics research. A high-fidelity simulation model of the BlueROV2 is implemented in Gazebo using existing ROS2-based simulation packages, enabling realistic modeling of vehicle dynamics, actuator behavior, buoyancy, hydrodynamic drag, and environmental interactions in a controlled and repeatable setting. This simulation framework provides a practical testbed for controller development without the cost and operational complexity associated with field experiments.

The objective of this work is to develop a nonlinear Model Predictive Control (NMPC) framework for waypoint tracking and depth regulation of the BlueROV2. The controller must generate appropriate generalized force and moment commands that drive the vehicle toward desired reference states while accounting for nonlinear system behavior and actuator constraints. At each control timestep, the controller solves a constrained finite-horizon optimization problem using a predictive model of the vehicle dynamics and applies the first control action in a receding horizon manner.

IV. METHOD

The proposed control framework combines a nonlinear dynamic model of the BlueROV2, a nonlinear Model Predictive Controller (NMPC), and a closed-loop simulation architecture implemented in ROS2 and Gazebo. The overall methodology is designed to enable accurate trajectory tracking and depth regulation while explicitly accounting for the coupled nonlinear dynamics and actuator limitations of the underwater vehicle.

The framework operates in a receding-horizon closed-loop configuration, as shown in Figure 1. First, a desired waypoint or reference trajectory is provided in the form of a target state vector containing position and heading references. At each control timestep, the current vehicle state is estimated from odometry feedback obtained through the Gazebo simulation environment. This state estimate includes translational position, orientation, linear velocity, and angular velocity, and is used to initialize the controller.

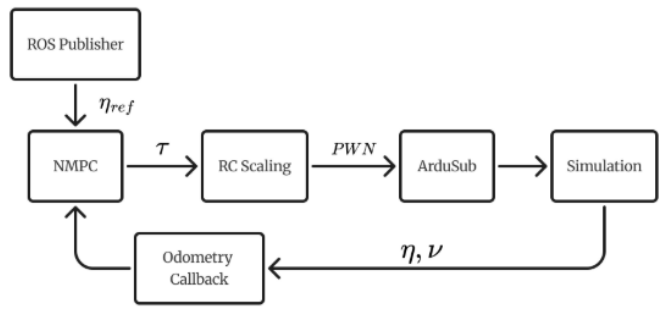

Fig. 1. Closed-loop NMPC control architecture for BlueROV2 simulation.

The NMPC then solves a constrained finite-horizon optimization problem using the nonlinear vehicle model as its prediction model. The optimization computes the generalized control input vector τ which consists of desired body-frame forces and moments in surge, sway, heave, roll, pitch, and yaw. The controller minimizes a quadratic objective that penalizes state tracking error and excessive control effort while satisfying dynamic constraints and actuator limits.

The resulting optimal control action is converted into RC override commands compatible with the ArduSub flight controller. Because ArduSub performs internal thruster mixing and low-level stabilization, the NMPC outputs are treated as generalized force and moment commands rather than individual thruster commands. These commands are independently scaled and mapped into PWM control signals, which are transmitted through MAVROS to the software-in-the-loop (SITL) vehicle controller.

Finally, ArduSub converts the PWM override inputs into motor-level actuation commands for the simulated BlueROV2 in Gazebo. The vehicle response is then propagated through the nonlinear hydrodynamic simulation, producing updated state feedback that is returned to the controller for the next optimization step. This repeated feedback-optimization-actuation loop forms a closed-loop predictive control architecture capable of continuously correcting trajectory deviations and compensating for disturbances in real time.

This methodology provides a systematic framework for integrating model-based nonlinear control with high-fidelity underwater simulation, enabling repeatable evaluation of trajectory tracking performance, control effort, and robustness under realistic operating conditions.

V. MATHEMATICAL MODEL

This section presents the mathematical formulation of the underwater vehicle dynamics using a 6-DOF rigid-body model with hydrodynamic effects. The model follows the standard marine vehicle notation introduced by Fossen [10].

The vehicle state is defined in the world frame and body-fixed frame as:

$$\eta = [x, y, z, \phi, \theta, \psi]^T$$

$$\nu = [u, v, w, p, q, r]^T$$

$$\tau = [X, Y, Z, K, M, N]^T$$

where η represents position and orientation, ν represents linear and angular velocities in the body frame, and τ represents generalized forces and moments. The coordinate frames used are shown in Figure 2:

The model uses the following assumptions:

1. Center of buoyancy is located at the origin: $(0, 0, 0)$.
2. Center of gravity is located at $(0, 0, z_g)$.
3. Inertia cross-coupling terms are negligible: $I_{xy} = I_{xz} = I_{yz} = 0$.
4. The vehicle is neutrally buoyant: $mg = \rho g V$.
5. Ocean currents are constant.
6. Thrusters are operating at 16V.

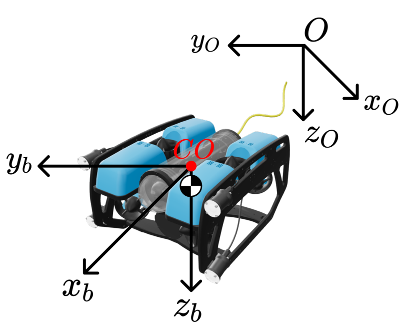

Fig. 2. Inertial frame and body frame of the BlueROV.

A. Kinematics

The kinematic relationship between the inertial frame and body frame is given by [5], [11]:

$$\dot{\eta} = J(\eta)\nu \tag{1}$$

where the transformation matrix $J(\eta)$ is:

$$J(\eta) = \begin{bmatrix} R(\Theta) & 0_{3\times3} \\ 0_{3\times3} & T(\Theta) \end{bmatrix} \tag{2}$$

with $\Theta = [\phi, \theta, \psi]^T$.

The rotation matrix $R(\Theta)$ maps body-frame velocities to the inertial frame, while $T(\Theta)$ relates angular velocities to Euler angle rates:

$$R(\Theta) \in SO(3), \quad T(\Theta) \in R^{3\times3}$$

where $s(\cdot) = \sin(\cdot)$, $c(\cdot) = \cos(\cdot)$, and $t(\cdot) = \tan(\cdot)$.

B. Kinetics

The dynamics of the underwater vehicle are governed by rigid-body and hydrodynamic effects.

The total inertia matrix is composed of rigid-body and added-mass components:

$$M = M_{RB} + M_A \tag{3}$$

The rigid-body inertia matrix is:

$$M_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & 0 \\ 0 & m & 0 & -mz_g & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & -mz_g & 0 & I_x & 0 & 0 \\ mz_g & 0 & 0 & 0 & I_y & 0 \\ 0 & 0 & 0 & 0 & 0 & I_z \end{bmatrix} \tag{4}$$

The added mass matrix is:

$$M_A = -\text{diag}(X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}) \tag{5}$$

The Coriolis matrix includes rigid-body and added-mass contributions:

$$C(\nu) = C_{RB} + C_A \tag{6}$$

where C_{RB} arises from rigid-body motion and C_A from hydrodynamic added mass effects.

The damping forces are modeled as linear and quadratic contributions [12]:

$$D(\nu) = -\text{diag}\Big(X_u + X_{u|u|}|u|, Y_v + Y_{v|v|}|v|, Z_w + Z_{w|w|}|w|, K_p + K_{p|p|}|p|, M_q + M_{q|q|}|q|, N_r + N_{r|r|}|r|\Big) \tag{7}$$

The restoring force vector due to gravity and buoyancy is:

$$g(\eta) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ z_g W \cos\theta \sin\phi \\ z_g W \sin\theta \\ 0 \end{bmatrix} \tag{8}$$

C. Equation of Motion

The full nonlinear 6-DOF equations of motion are [13]:

$$\dot{\eta} = J(\eta)\nu \tag{9}$$

$$M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau \tag{10}$$

D. Thrust to PWM Mapping

The BlueROV2 receives actuator commands through the `/rc/override` interface, which accepts PWM signals rather than direct force or moment inputs. As a result, the controller output is converted into PWM commands centered at 1500 μs, where values above and below this neutral point generate positive and negative thrust, respectively. Assuming thruster operation at 16V, corresponding to the nominal operating condition, the thrust–PWM relationship is derived using Blue Robotics T200 thruster performance data [14] and curve-fitted, as shown in Figure 3. This calibrated mapping enables the desired control inputs to be translated into actuator commands while incorporating deadband and saturation effects.

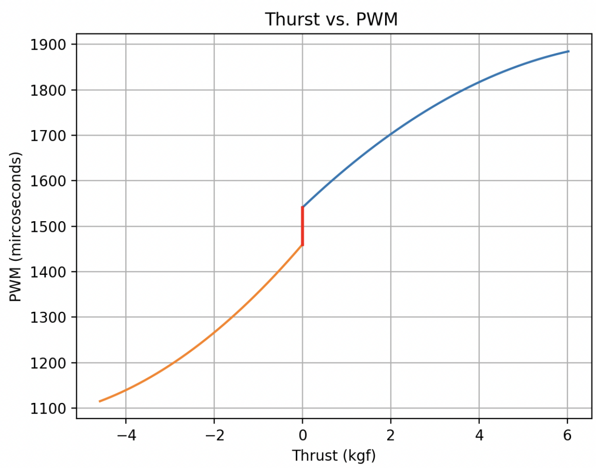

Fig. 3. Thrust To PWM Curve

VI. CONTROL SYSTEM

The NMPC minimizes a quadratic cost over a finite prediction horizon, consisting of stage and terminal components. The stage cost penalizes both state tracking error and control effort, while the terminal cost enforces convergence to the reference state.

$$\begin{aligned}\min_{\{u_k\}_{k=0}^{N-1}} \quad J = \sum_{k=0}^{N-1} \Big[&(x_k - x_{\mathrm{ref}})^T Q (x_k - x_{\mathrm{ref}}) \\ &+ u_k^T R u_k \Big] \\ + (x_N - x_{\mathrm{ref}})^T &Q_f (x_N - x_{\mathrm{ref}})\end{aligned} \tag{11}$$

The state weighting matrix Q is selected to prioritize positional accuracy by assigning larger weights to the translational states (x, y, z) while comparatively smaller weights are assigned to the orientation states to permit moderate attitude deviations. The input weighting matrix R regularized control effort, promoting smooth and physically realizable actuation.

At each sampling instant, the NMPC solves a constrained nonlinear optimization problem to compute an optimal control sequence over the prediction horizon. The optimization is performed subject to the system dynamics and the imposed input and state constraints.

A receding horizon strategy is employed, wherein only the first control input of the optimal sequence is applied to the system. The optimization is then repeated at the next timestep using updated state measurements, providing closed-loop feedback and robustness to disturbances and modeling errors.

A. Implementation

The controller is implemented using the `do-mpc` Python library, which provides a high-level interface for nonlinear MPC design. The system dynamics are defined symbolically using CasADi, enabling efficient automatic differentiation and numerical optimization. A direct collocation method is used for discretization of the continuous-time dynamics, allowing accurate handling of nonlinearities within the prediction horizon.

The NMPC controller is integrated into the Gazebo simulation through the Robot Operating System (ROS). A dedicated ROS node is developed to run the controller, which subscribes to vehicle state estimates published by the simulation. These states are used as inputs to the NMPC optimization problem at each control step. The computed control inputs are then published as thruster commands to the appropriate ROS topics, which interface with the BlueROV2 model in Gazebo via MAVROS.

VII. SIMULATION

The simulation environment is based on the `orca4` [9] ROS2 framework, which provides a high-fidelity model of the BlueROV2 autonomous underwater vehicle (AUV). The framework integrates Gazebo Harmonic for physics-based simulation, ArduSub for low-level vehicle control, and ROS2 for system-level communication and coordination.

The vehicle dynamics are simulated using standard hydrodynamic and buoyancy plugins within Gazebo, capturing key effects such as added mass, drag, and restoring forces. The simulated environment consists of a water surface at $z = 0$ and a flat seafloor located approximately 10 meters below, providing a representative operating depth for subsea tasks.

Simulation is executed using a software-in-the-loop (SITL) configuration, where ArduSub communicates with Gazebo through the `ardupilot_gazebo` interface. This setup enables realistic testing of control algorithms by incorporating actuator dynamics and sensor feedback within a closed-loop environment.

VIII. RESULTS

A position-holding test is performed at the reference state

$$\eta_{ref} = [0, 0, -1, 0, 0, 0]$$

for 140 seconds with the first 5 seconds of settling time cut. This test evaluates the controller's ability to regulate the vehicle state under closed-loop conditions while minimizing steady-state error and excessive control effort.

Figure 4 shows the vehicle trajectory during the position-hold maneuver. Figure 5 presents the state tracking error over time, and Figure 6 shows the corresponding control inputs generated by the NMPC.

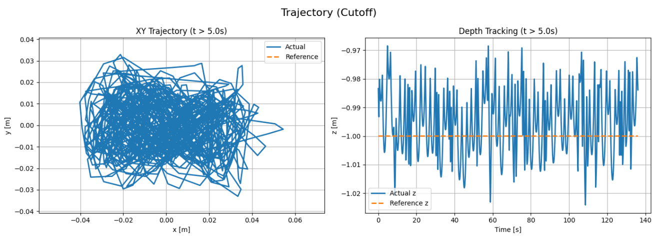

Fig. 4. Vehicle trajectory during position-hold test.

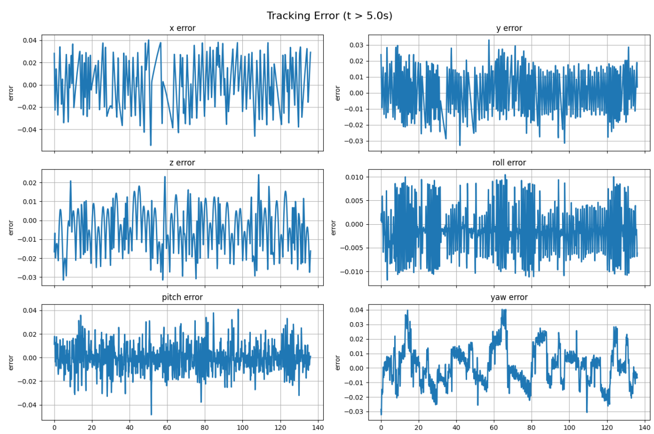

Fig. 5. Tracking error during position-hold test.

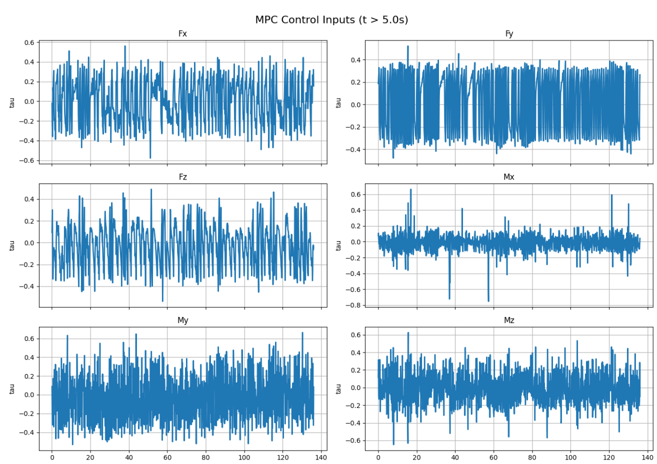

Fig. 6. NMPC control inputs during position-hold test.

The latter test is performed over a waypoint-tracking mission with the following reference states:

TABLE I

POSITION-HOLD PERFORMANCE METRICS.

	RMSE Value
x	0.0192 m
y	0.0120 m
z	0.0117 m
θ	0.0112 rad
ψ	0.0131 rad

$$\eta_{ref,0} = [0, 0, 0, 0, 0, 0]$$
$$\eta_{ref,1} = [4, 4, -4, 0, 0, 0]$$
$$\eta_{ref,2} = [5, 2.5, -5, 0, 0, 0]$$
$$\eta_{ref,3} = [-3, 0, -5, 0, 0, -0.5]$$
$$\eta_{ref,4} = [-4, -5.5, -5, 0, 0, 0]$$
$$\eta_{ref,5} = [8, -2.5, -7, 0, 0, 0]$$

This trajectory evaluates the NMPC's ability to track multiple setpoints with varying position, depth, and heading references under closed-loop operation. Figure 7 shows the resulting vehicle trajectory. Figure 8 presents the corresponding tracking error, while Figure 9 shows the control effort required throughout the maneuver.

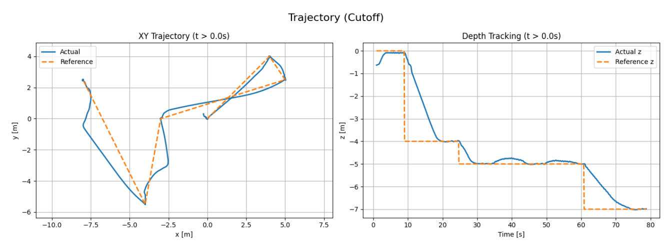

Fig. 7. Waypoint-tracking trajectory under NMPC control.

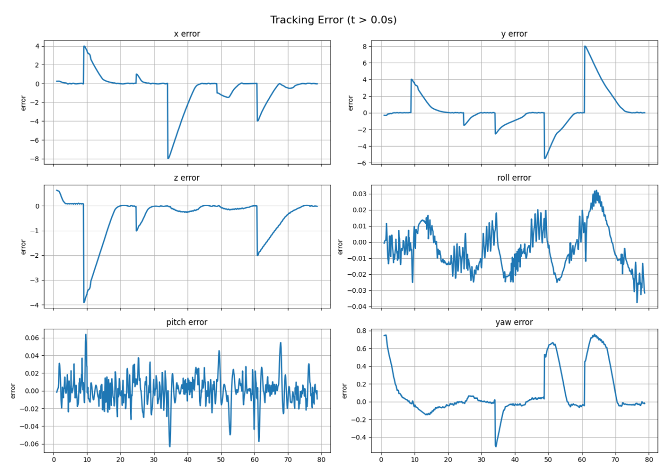

Fig. 8. State tracking error during waypoint-tracking test.

IX. PERFORMANCE EVALUATION

The results indicate that the NMPC controller provides stable closed-loop regulation and waypoint-tracking behavior

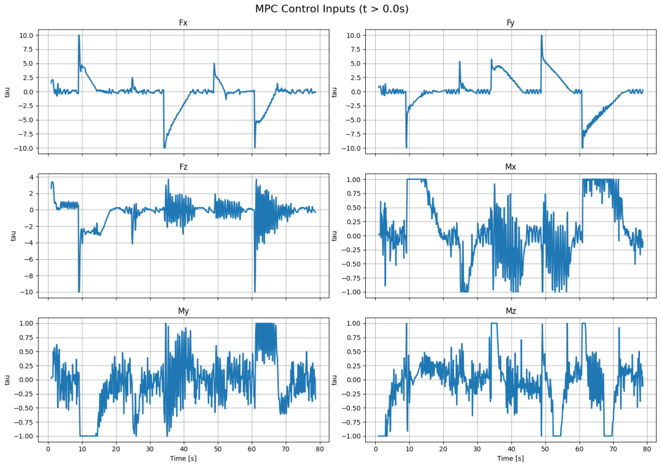

Fig. 9. Control inputs generated during waypoint tracking.

in the BlueROV2 simulation. Since the numerical RMSE values for the position-holding case are reported in Table I, this section focuses on interpreting these results and comparing the observed behavior with previous studies of the BlueROV2 controller.

For the waypoint-tracking mission, the vehicle successfully transitioned between all reference waypoints while maintaining stable depth and heading regulation. Figure 7 shows that the vehicle follows the commanded trajectory with relatively small deviation despite the nonlinear coupled dynamics of the underwater system. The tracking error plots in Figure 8 indicate transient increases during aggressive waypoint transitions, followed by convergence toward the desired reference states. This behavior is expected because the controller must balance the accuracy of the trajectory with the actuator and dynamic constraints over the finite prediction horizon.

Table II compares the tracking performance of the proposed NMPC framework with previously reported BlueROV2 and underwater vehicle controllers from the literature. Previous studies using PID and LQR controllers report larger tracking errors, particularly in depth regulation, due to the limited ability of linear or purely feedback-based methods to handle nonlinear hydrodynamic coupling and actuator constraints. Similarly, sliding-mode approaches improve robustness but may introduce chattering and discontinuous control behavior. Compared to these methods, MPC-based controllers generally achieve lower RMSE values by explicitly optimizing future system behavior while enforcing system constraints.

Although the present work does not directly replicate the exact experimental conditions of the referenced studies, the obtained tracking performance is consistent with the expected advantages of nonlinear MPC for underwater vehicle control. In particular, the predictive optimization framework enables smoother control action, reduced steady-state tracking error, and improved handling of coupled multivariable dynamics. These results suggest that NMPC is a suitable control strategy for autonomous underwater navigation and waypoint-tracking tasks in realistic simulation environments.

TABLE II

RMSE COMPARISON OF UNDERWATER VEHICLE CONTROL STRATEGIES.

Controller	RMSE x [m]	RMSE y [m]	RMSE z [m]	Ref.
PID	0.036	0.024	0.034	[15]
LQR	0.011	0.008	0.086	
MPC	0.004	0.0076	0.0002	
PID	–	–	0.0407	[16]
SMC	–	–	0.0168	
SMC-FT	–	–	0.013	
PID	0.50 ± 0.14	0.58 ± 0.03	0.33 ± 0.12	[17]
REEF DRL	0.46 ± 0.13	0.59 ± 0.03	0.33 ± 0.09	
REEF-DR DRL	0.45 ± 0.14	0.55 ± 0.05	0.36 ± 0.15	
ABSC	0.1225	0.0597	0.1053	[18]
NLABSC	0.0985	0.0377	0.0764	

X. FUTURE WORK

For future development, an extension of this topic will be the addition of pipeline following and pipe damage detection [7] [19] [20], allowing the proposed system to autonomously detect and classify structural anomalies such as cracks, corrosion, and surface degradation along subsea pipelines.

Another direction is the integration of underwater mapping capabilities within the inspection framework. By incorporating techniques such as visual or sonar-based Simultaneous Localization and Mapping (SLAM), the vehicle could construct a 3D representation of the surrounding seabed while performing pipeline inspection. This would improve localization in GPS-denied environments and provide additional spatial context for the detected infrastructure.

Additionally, there is the possibility of exploring more realistic tether modeling for tethered AUV systems such as the BlueROV2. Recent work suggests a hybrid analytical-physical tether model in Gazebo where the cable is presented as a chain of buoyant segments connected by joints, allowing the tether to deform realistically under gravity, buoyancy, currents, and vehicle motion [21]. Environmental forces such as hydrodynamic drag and ocean currents act on each segment, allowing more accurate simulation of tether dynamics and vehicle interactions. Integrating this tether model into the simulation framework would improve the realism of AUV pipeline inspections and allow better evaluation of control strategies under more realistic marine conditions.

XI. CONCLUSIONS

This work presented a nonlinear Model Predictive Control framework for trajectory tracking and depth regulation of the BlueROV2 underwater vehicle within a high-fidelity ROS2 and Gazebo simulation environment. The controller incorporated nonlinear vehicle dynamics and actuator constraints into a finite-horizon optimization problem solved in a receding-horizon manner.

Simulation results demonstrated stable closed-loop behavior, accurate position holding, and effective waypoint tracking with low steady-state error. The controller also produced smooth control inputs while handling the coupled nonlinear dynamics of the underwater vehicle.

Overall, the results indicate that NMPC is a promising control strategy for autonomous underwater vehicle navigation and provide a foundation for future work in underwater inspection, mapping, and autonomous operation.

APPENDIX

GitHub Repository: `https://github.com/maggien2/bluerov_mpc`

TABLE III
NOMENCLATURE USED IN THE DYNAMIC MODEL.

Symbol	Description
X_u, Y_v, Z_w	Linear damping coefficients for translational motion in water
K_p, M_q, N_r	Linear damping coefficients for rotational motion in water
$X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}$	Added mass terms for translational motion in water
$K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}$	Added inertia terms for rotational motion in water
$X_{u\|u\|}, Y_{v\|v\|}, Z_{w\|w\|}$	Quadratic damping coefficients for translational motion in water
$K_{p\|p\|}, M_{q\|q\|}, N_{r\|r\|}$	Quadratic damping coefficients for rotational motion in water
W, B	Weight and buoyant force
m	Mass of the ROV
CO	Center of Origin
PWM	Pulse-Width Modulation
SMC	Sliding Mode Controller
SMC-FT	SMC Finite Time
REEF DRL	Robust and Energy Efficient Framework Deep Reinforcement Learning
REEF-DR DRL	REEF DRL with domain randomization
ABSC	Adaptive Backstepping Controller
NLABSC	Nonlinear ABSC

TABLE IV
ESTIMATED COEFFICIENTS [5] [12].

Symbol	Value	Unit
m	11.4	kg
I_x	0.21	kgm^2
I_y	0.245	kgm^2
I_z	0.245	kgm^2
g	9.82	kg/m^2
z_g	0.02	m
X_u	13.7	Ns/m
$X_{u\|u\|}$	141.0	Ns^2/m^2
Y_v	0	Ns/m
$Y_{v\|v\|}$	217.0	Ns^2/m^2
Z_w	33.0	Ns/m
$Z_{w\|w\|}$	190.0	Ns^2/m^2
K_p	0	Ns/m
$K_{p\|p\|}$	1.19	Ns^2/m^2
M_q	0.8	Ns
$M_{q\|q\|}$	0.47	Ns^2
N_r	0	Ns
$N_{r\|r\|}$	1.5	Ns^2
$X_{\dot{u}}$	6.36	kg
$Y_{\dot{v}}$	7.12	kg
$Z_{\dot{w}}$	18.68	kg
$K_{\dot{p}}$	0.189	kgm^2
$M_{\dot{q}}$	0.135	kgm^2
$N_{\dot{r}}$	0.222	kgm^2

ACKNOWLEDGMENT

LLMs were used to assist with grammar correction and preliminary literature search.

REFERENCES

[1] M. Sørensen, F. F. Uth, E. T. Jouffroy, J. . Liniger, and J. Pedersen, "An Open-Source Benchmark Simulator: Control of a BlueROV2 Underwater Robot," *Journal of Marine Science and Engineering*, vol. 10, no. 12, p. 1898, 2022. [Online]. Available: https://www.mdpi.com/2077-1312/10/12/1898

[2] W. Akram and A. Casavola, "A visual control scheme for AUV underwater pipeline tracking," *ICAS 2021 - 2021 IEEE International Conference on Autonomous Systems, Proceedings*, 8 2021. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9551173

[3] T. Xia, D. Cui, Z. Chu, and X. Yu, "Autonomous Heading Planning and Control Method of Unmanned Underwater Vehicles for Tunnel Detection," *Journal of Marine Science and Engineering*, vol. 11, no. 4, 4 2023. [Online]. Available: https://www.mdpi.com/2077-1312/11/4/740

[4] C. Shen, Y. Shi, and B. Buckham, "Model predictive control for an AUV with dynamic path planning," *2015 54th Annual Conference of the Society of Instrument and Control Engineers of Japan, SICE 2015*, pp. 475–480, 9 2015. [Online]. Available: https://ieeexplore.ieee.org/document/7285374

[5] E. M. Einarsson and A. Lipenitis, "Model Predictive Control for the BlueROV2 Theory and Implementation," *Aalborg Universitet*, 10 2020. [Online]. Available: https://projekter.aau.dk/projekter/files/387609647

[6] M. M. M. Manhães, S. A. Scherer, M. Voss, L. R. Douat, and T. Rauschenbach, "UUV Simulator: A Gazebo-based package for underwater intervention and multi-robot simulation," *OCEANS 2016 MTS/IEEE Monterey, OCE 2016*, 11 2016. [Online]. Available: https://ieeexplore.ieee.org/document/7761080

[7] I. C. Sang and W. R. Norris, "An Autonomous Underwater Vehicle Simulation With Fuzzy Sensor Fusion for Pipeline Inspection," *IEEE Sensors Journal*, vol. 23, no. 8, pp. 8941–8951, 4 2023. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10061378

[8] A. Agrawal, E. Palmer, Z. Kingston, and G. A. Hollinger, "Underwater Multi-Robot Simulation and Motion Planning in Angler," 6 2025. [Online]. Available: https://arxiv.org/pdf/2506.06612v1http://dx.doi.org/10.1109/OCEANS58557.2025.11104649

[9] C. McQueen, "clydemcqueen/orca4: Ros2 auv based on the bluerov2, ardusub and navigation2," 2025. [Online]. Available: https://github.com/clydemcqueen/orca4?tab=readme-ov-file

[10] T. I. Fossen, "Handbook of marine craft hydrodynamics and motion control," *Handbook of Marine Craft Hydrodynamics and Motion Control*, 4 2011. [Online]. Available: https://onlinelibrary.wiley.com/doi/book/10.1002/9781119994138

[11] H. T. Nguyen, V. Q. Nguyen, H. Quan Vo, V. T. Ngo, and D. T. Tran, "IOP Conference Series: Earth and Environmental Science LQR-Based Tracking Control for Remotely Operated Vehicles: A Co-Simulation Framework for Sustainable Environmental Monitoring LQR-Based Tracking Control for Remotely Operated Vehicles: A Co-Simulatio. . . ."

[12] M. von Benzon, F. F. Sørensen, E. Uth, J. Jouffroy, J. Liniger, and S. Pedersen, "An Open-Source Benchmark Simulator: Control of a BlueROV2 Underwater Robot," *Journal of Marine Science and Engineering 2022, Vol. 10,*, vol. 10, no. 12, 12 2022. [Online]. Available: https://www.mdpi.com/2077-1312/10/12/1898

[13] C.-J. Wu, "6-DoF Modelling and Control of a Remotely Operated Vehicle," 2018.

[14] [Online]. Available: https://bluerobotics.com/store/thrusters/t100-t200-thrusters/t200-thruster-r2-rp/(cit.onpp.3,25).

[15] Y. Cao, B. Li, Q. Li, A. A. Stokes, D. M. Ingram, and A. Kiprakis, "A nonlinear model predictive controller for remotely operated underwater vehicles with disturbance rejection," *IEEE Access*, vol. 8, pp. 158 622–158 634, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9180253

[16] J. González-García, A. Gómez-Espinosa, L. G. García-Valdovinos, T. Salgado-Jiménez, E. Cuan-Urquizo, and J. A. E. Cabello, "Experimental validation of a model-free high-order sliding mode controller with finite-time convergence for trajectory tracking of autonomous underwater vehicles," *Sensors (Basel, Switzerland)*, vol. 22, p. 488, 1 2022. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8780515/

[17] V. Sufan and G. Troni, "Swim4real: Deep reinforcement learning-based energy-efficient and agile 6-dof control for underwater vehicles," *IEEE Robotics and Automation Letters*, vol. 10, pp. 7326–7333, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11020757

[18] S. Eddine Seghiri, A. Chemori, N. Mansouri, and V. Creuze, “A novel adaptive approach to underwater ROV control: design and real-time experiments A novel adaptive approach to underwater ROV control: Design & real-time experiments,” *Journal of Marine Science and Technology*, vol. 30, 2025. [Online]. Available: https://hal-lirmm.ccsd.cnrs.fr/lirmm-04982014v1

[19] S. K. Kartal and R. F. Cantekin, “Autonomous Underwater Pipe Damage Detection Positioning and Pipe Line Tracking Experiment with Unmanned Underwater Vehicle,” *Journal of Marine Science and Engineering 2024, Vol. 12,*, vol. 12, no. 11, 11 2024. [Online]. Available: https://www.mdpi.com/2077-1312/12/11/2002

[20] H. Zhang, S. Zhang, Y. Wang, Y. Liu, Y. Yang, T. Zhou, and H. Bian, “Subsea pipeline leak inspection by autonomous underwater vehicle,” *Applied Ocean Research*, vol. 107, 2 2021.

[21] M. Buchholz, I. Carlucho, M. Grimaldi, and Y. R. Petillot, “Tethered Multi-Robot Systems in Marine Environments,” 8 2025. [Online]. Available: https://arxiv.org/pdf/2508.02264v1